

Name:

ID:

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[Points] **Question 1:** Determine whether the following sentences are **TRUE** or **FALSE**.
Correct the **FALSE** statements.

1. (T / F) Floating-point values are represented as fraction and exponent.
2. (T / F) Type checking can be done statically and in the same time type binding can be done dynamically.

[6.5 Points] **Question 2:** Choose the **correct answer** for each of the following statements:

- 1) _____ is a collection of the attributes of a variable.
 - a. Data type
 - b. Object
 - c. Descriptor
 - d. Scope
 - e. None of the above
- 2) Which of the following languages which use level numbers in record declaration:
 - a. C++
 - b. Ada
 - c. Python
 - d. COBOL
 - e. None of the above
- 3) Which of the following is **not** an ordinal type:
 - a. Enumeration
 - b. Subrange
 - c. Boolean
 - d. Char
 - e. None of the above
- 4) The address of element A[1,2] where A is a two-dimensional array with subscript limits $1 \leq i \leq 6$ and $1 \leq j \leq 7$ and A is in row major addressing. Assume base address is 1000 and word size is 2:
 - a. 1016
 - b. 1014
 - c. 1002
 - d. 1012
 - e. None of the above
- 5) Jagged array is a multidimensional array in which:
 - a. All rows have same number of elements.
 - b. The lengths of the rows need not to be the same.
 - c. The elements need not to be the same type.
 - d. Array elements are mapped to one-dimension array in row major order.
 - e. None of the above
- 6) Which of the following data types **cannot** used as array index:
 - a. Boolean
 - b. Float
 - c. Character
 - d. Negative integer
 - e. None of the above

- 7) Which of the following is **not true** about Associative arrays:
- Associative array is an unordered collection of data elements.
 - The size of an associative array is static.
 - In searching, associative arrays are better than traditional arrays.
 - None of the above
- 8) Decimal data type is designed specially to support:
- Scientific applications
 - Business applications
 - Artificial intelligent applications
 - All of the above
 - None of the above

Language	Expression	Value
Ada	-9 mod 4	
Ada	3**1**2**-5	
APL	2*3+5	
C language	Int x= 5>7;	
VB	2^0^5	
C language	int x=15; int y = (x=5)?3:4;	
C sharp	int x=15; int y=5; (x>10)?y :x=20 ;	
C language	5+1<6+2<7+2	
Java	int x=(2 -2 +5 -5)*(210/3*9-201)	
Pascal	Bolean x ; x := 5=1 ;	
C language	int a = 3 ; int x = !a ;	
PHP	"91" === 91	